



Simplified absorption line selection methods for water absorption thermometry around 2 μm

Mengmeng Tao^{a,b,*}, Yamin Wang^a, Haolong Wu^a, Jingfeng Ye^a, Xisheng Ye^b, Weibiao Chen^b

^a State Key Laboratory of Laser Interaction with Matter, Northwest Institute of Nuclear Technology, Xi'an 710024, China

^b Wang Zhijiang Laser Innovation Center, Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghai 201800, China



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ABSTRACT

Line-selection is of vital importance for absorption spectroscopy based thermometry. Up to now, various analytical and numerical line-selection methods have been developed to tackle this issue. Here, concerning the numerical line-selection of water around 2 μm , the conventional method based on the frequency-dependent lower-state energy is simplified through pre-filtering of the candidate absorption lines. As a result, the previous thousands of candidate lines is greatly reduced to a few dozen. With reduced libraries, three simplified line-selection methods are proposed. Line selection results of the simplified methods in the investigated wavelength span from 1800 nm to 1900 nm are discussed and evaluated covering a wide temperature range from 400 K to 2000 K. Comparisons with the conventional method demonstrates that, these three simplified methods provide comparable temperature precision, but with a much faster selection speed.

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1. Introduction

Molecular absorption spectroscopy has been widely exploited in various fields, including chemical analysis [1–4], combustion diagnosis [5–7], breath monitoring [8–11], health testing and diagnosis [12–16]. And, most of these applications involves the measurement of temperature, i.e., absorption based thermometry. For molecular absorption based thermometry, usually, two absorption lines are monitored, and the temperature is inferred from the ratio of the two measured absorbance [16–26]. This makes the selection of optimal absorption transitions the first and foremost step in designing an absorption based temperature sensor. And, it has been widely recognized that, the selection of the absorption lines is critical to the design of a sensitive, reliable and robust temperature sensor, as it directly affects the accuracy, complexity and cost of the system [16,19–24].

Heretofore, enormous efforts have been devoted to this issue and various wavelength selection strategies have been developed concerning different application scenes. Generally, these methods can be divided into two categories. The first category features the development of a systematic line-selection procedure containing 5 to 6 analytical criteria, of which absorption strength, temperature sensitivity and spectral isolation from other species are major con-

cerns. In 1997, for water vapor based thermometry in high pressure combustion gases, V. Nagali proposed 5 selection rules with special emphasis on the maximization of temperature sensitivity [25]. With these principles, tens of optimal wavelength pairs are suggested in 1.3–1.4 μm . Later, concerning the temperature measurement with a single tunable diode laser, X. Zhou brought forward a set of 6 norms with more specific requirements for wavelength selection guiding [19,20,26]. With the recommended design rules, 10 optimal wavelength pairs around 1.9 μm and 12 optimal wavelength pairs spanning over 1.3–2.0 μm are selected. Thereafter, this analytical line-selection procedure has been widely accepted for line pair selection in two-color thermometry [16,21–24].

However, wavelength selection is still not an easy task, as with these combined criteria which are mostly qualitative descriptions, it is not entirely obvious how to effectively choose lines to guarantee optimal performance at given temperatures. Firstly, these criteria are derived for the relatively simple case of two-color thermometry, which might not be applicable for more complicated cases encountering multi-line absorption. Besides, for two-color thermometry, the line pairs selected are usually required to be free of significant interference from nearby transitions [20–24], which is hardly possible for water in high pressure environments where significant overlapping and blending is prominent.

The second category focuses on the numeralization of the line-selection process, which converts the analytical criteria into a mathematical problem, an optimization problem, to be specific. In 2009, A. W. Caswell proposed a new quantity named frequency-

* Corresponding author at: State Key Laboratory of Laser Interaction with Matter, Northwest Institute of Nuclear Technology, Xi'an 710024, China.

E-mail address: tonylemon@yeah.net (M. Tao).

dependent lower-state energy E_v , which presents the weighted sum of all the lower-state energies that contribute to the absorption at a particular wavelength [27]. Later, the properties of E_v were investigated in detail by X. An [28]. Studies show that, different from the conventional lower-state energy E_0 , which is a discrete constant for specific absorption line, E_v is a continuous quantity and varies with encountered environmental conditions. Quantitatively, E_v can be easily calculated with the spectral ratio at two different temperatures. And, the temperature uncertainty of selected lines can be derived through E_v . Thus, mathematically, the complicated line-selection problem is converted into an optimization problem, which is a quantifiable and numerical process rather than a qualitative and analytical one. Exploiting the frequency-dependent lower-state energy, various wavelength selection problems have been effectively resolved featuring different wavelength numbers and different temperature conditions for water absorption thermometry around 1.3–1.4 μm [29–32].

In Caswell and An's method (CA method), the lines are selected through the brute-force algorithm or the differential evolution algorithm. In brute-force algorithm, to find the best absorption lines, all possible wavelength combinations have to be tested thoroughly, bringing a tremendous amount of computation. Compared with brute-force, differential evolution algorithm, designed to resolve optimization problems, can greatly improve the line-selection efficiency [32–34]. Yet, the line-selection process is still time consuming, especially for multi-line-selection cases, due to a huge sample space containing a large amount of absorption lines.

Extensive studies have shown that the 2 μm region holds huge potential for absorption spectroscopy application, as it possesses more abundant and much stronger H_2O and CO_2 absorption lines compared to the widely exploited telecommunication window [19,26,35–37]. Recently, we investigated the hyperspectral absorption characteristics of room-temperature water vapor around 2 μm with a self-developed widely tunable, narrow linewidth Tm-doped fiber laser [38,39]. Detailed absorption peaks are detected and well-resolved with the measured absorption coefficients agree well with the theoretical simulation quantitatively. Later, utilizing the hyperspectral absorption of water around 2 μm , the temperature and water mole fraction of air and alcohol flame are derived through comparisons with the HITRAN2016 database [40]. However, in previous researches, the hyperspectral measurements were conducted without careful line-selections, but with widely tuning laser sources covering a spectral range over 40 nm. Data recording covering such a wide spectral range poses a severe challenge to the data acquisition and processing system at high repetition rates. A proper solution involves the selection of several absorption lines or bands to reduce the data volume as demonstrated in [41].

Here, concerning the absorption line selection of water around 2 μm , three simplified line-selection methods are proposed taking advantage of the frequency-dependent lower-state energy concept. The basic ideas and line-selection processes of these methods are introduced. Lines selected by different methods are contrasted and evaluated. Comparisons with the traditional CA method show that, these three simplified methods feature comparable temperature precision and a much faster selection speed.

2. Methodology

2.1. CA method

2.1.1. Basic ideas

In Caswell and An's method, line-selection is realized based on the frequency-dependent lower-state energy E_v , which is derived

as [28]

$$E_v = \frac{k_B \ln[R_{T,T_0}(v) \sqrt{\frac{T}{T_0} \frac{Q(T)}{Q(T_0)}}]}{hc(\frac{1}{T_0} - \frac{1}{T})}, \quad (1)$$

where k_B is the Boltzmann's constant, h is the Planck's constant, c is the speed of light in vacuum. $Q(T)$ represents the partition function of water at temperature T , v is the wavenumber, and T_0 is the reference temperature. $R_{T,T_0}(v)$ denotes the absorption ratio, which can be written as

$$R_{T,T_0}(v) = \frac{a(T, v)}{a(T_0, v)} = \frac{\sum S(T) \phi(T, P, X, v)}{\sum S(T_0) \phi(T_0, P, X, v)}. \quad (2)$$

Here, $S(T)$ is the line strength (cm) and $\phi(T, P, X, v)$ is the line shape function (cm). Proportional to the logarithm of the absorption ratio $R_{T,T_0}(v)$, in fact, the frequency-dependent lower-state energy E_v represents the temperature sensitivity of absorption lines. A hot-line with an absorption ratio bigger than one corresponds to a large E_v , while, a cold-line with an absorption ratio less than one gives a small E_v .

In practical applications, with a denoted absorption measurement noise δa , the temperature uncertainty $\delta T/T$ can be expressed as [28]

$$\frac{\delta T}{T} = \frac{k_B T}{hc[(\sum E_v)^2 - N \sum E_v^2]} \sqrt{\sum \left(\frac{\delta a}{a}\right)^2 (\sum E_v - N E_v)^2}, \quad (3)$$

where N denotes the number of absorption lines selected within the investigated spectral range.

Since the favorable lines should provide an excellent temperature precision with low temperature uncertainty, to acquire the optimal absorption lines, we only need to find the proper combination of frequency-dependent lower-state energy E_v which corresponds to the minimum of Eq. (3). In this way, line-selection is transferred into an optimization problem which takes Eq. (3) as the objective function.

2.1.2. Selection process

Genetic algorithm (GA), based on natural selection, has been widely exploited as a useful tool for solving various optimization problems in spectral analysis [42–44]. Compared to the differential evolution algorithm, GA is more convenient and accessible, as the GA function can be directly called in some commercial software. Here, GA is utilized to find the optimal absorption lines in CA method with the selection process illustrated in Fig. 1.

As can be seen in Fig. 1, the whole selection process contains the following three steps:

- (1) Generation of the absorption line library. Here, the absorption line library is directly taken as the HITRAN2016 database.
- (2) Derivation of the objective function. The objective function is of vital importance as it defines the problem that needs to be optimized. Firstly, absorption spectra $a(T, v)$ and $a(T_0, v)$ are simulated at given conditions with HITRAN2016 database. Then, with the frequency-dependent lower-state energy E_v attained through Eq. (1), the objective function can be derived by Eq. (3).
- (3) Line-picking through GA optimization. GA approach is utilized to solve the optimization problem defined by the objective function with lines from the absorption line library. The optimization results represent the optimal absorption lines.

Usually, in the optimization process, an iterative GA loop is introduced to ensure a global optimal solution over the whole spectral range in consideration.

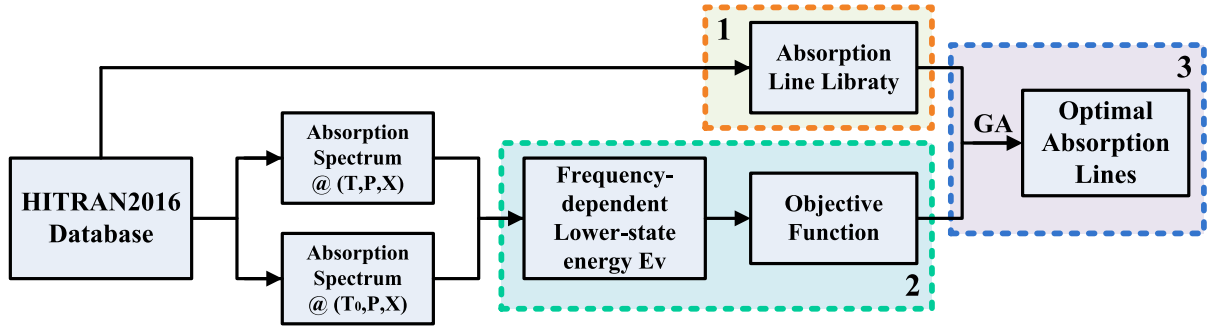


Fig. 1. Line-selection process with GA approach in CA method.

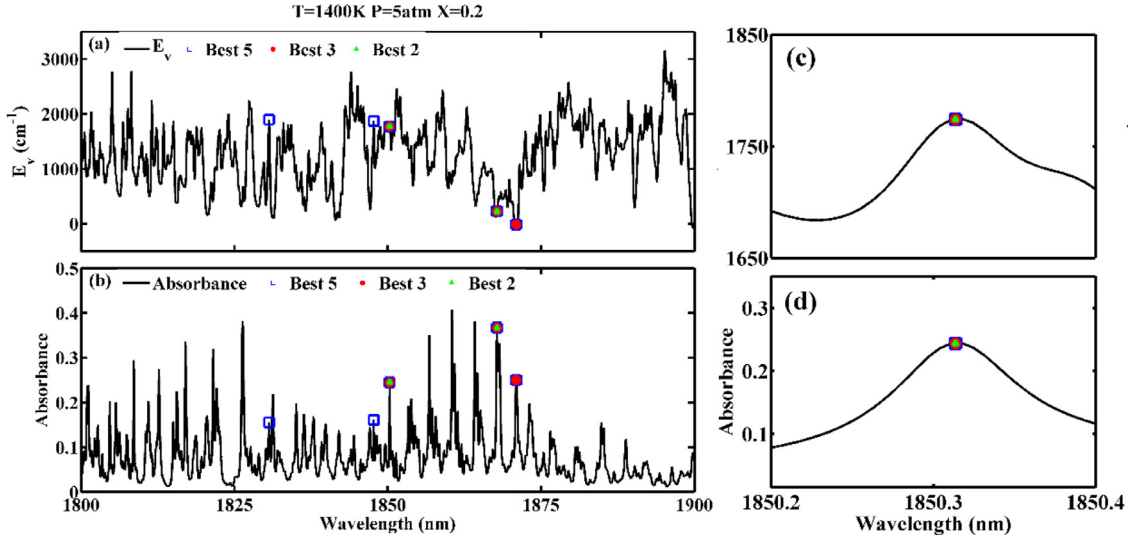


Fig. 2. Best selections for 2, 3 and 5 lines at 1400 K. (a) frequency-dependent lower-state energy E_v ; (b) absorbance; (c) and (d) are partial enlargements of (a) and (b) around 1850.3 nm. The green triangles, red dots and blue squares represent the best selections of 2, 3 and 5 lines, respectively.

2.1.3. Typical selection results

For the concerned spectral range from 1800 nm to 1900 nm, typical selection results for 2, 3 and 5 lines at $T = 1400\text{K}$ for given conditions are depicted in Fig. 2. Here, the reference temperature is set to be $T_0 = 0.99T$, while the absorption measurement noise is taken as $\delta a = 1 \times 10^{-4}$. And, these simulation conditions are used throughout this paper unless otherwise stated.

For the investigated wavelength range, there are about 3000 absorption lines in total given by the HITRAN2016 database. And, as can be found in Fig. 2, the best selections always lie at both the peaks of absorption and the extremes of the frequency-dependent lower-state energy E_v . Located at absorption peaks, these lines, with strong absorptions, help to reduce the noise induced uncertainty as inferred in Eq. (3), which is consistent with the findings in Ref. [30–32]. At the limits of the frequency-dependent lower-state energy E_v , these lines are the most sensitive ones to temperature changes in the investigated spectral range. Lines at energy peaks are named as hot-lines, while lines at energy valleys are labeled as cold-lines. It can be found in Fig. 2(a) that, the selected results contain at least one hot-line at the peak and one cold-line at the valley. For example, the best 2 lines consist one hot-line and one cold-line, while the best 5 lines consist 3 hot-lines and 2 cold-lines. And, this combination of hot-lines and cold-lines is favorable for the enhancement of the temperature measurement precision, as they provide large energy differences.

Besides, interestingly, the selection results show that the best $N + 1$ lines include the best N lines, which is beneficial for practical applications, as the sensor system reveals excellent compatibility and extendibility with the increase of line number.

2.2. Simplified method with a reduced library

The selection principles and results of different line-selection methods, including the analytical ones and the numerical ones, show that, favorable lines should satisfy two basic requirements, i.e. sufficient absorption strength and high temperature sensitivity. Thus, the CA method can be simplified by reducing the sample space through line-filtering based on the absorption strength and temperature sensitivity.

For any numerical line-selection method, first of all, the quantification of the absorption strength and the temperature sensitivity should be addressed. The absorption strength can be directly calculated with the HITRAN2016 database. And, the temperature sensitivity can be quantified through the frequency-dependent lower-state energy E_v . Then, lines with both strong absorptions and high temperature sensitivity can be picked to form the reduced candidate library. Since this simplified method is in fact the CA method with a reduced library, here, it's labeled as the CARL method. And, Fig. 3 illustrates the line-selection process for the proposed CARL method.

As can be found in Fig. 3, similar to the CA method given in Fig. 1, the line-selection process of CARL method also contains three major steps: generation of the reduced absorption line library, derivation of the objective function, and line-picking through GA optimization. The difference lies in the first step, while the second and the third steps are the same as those described in CA method. Instead of taking the HITRAN2016 database directly as the line library, the generation of the reduced library involves the establishment of two line collections, namely the strong-lines collec-

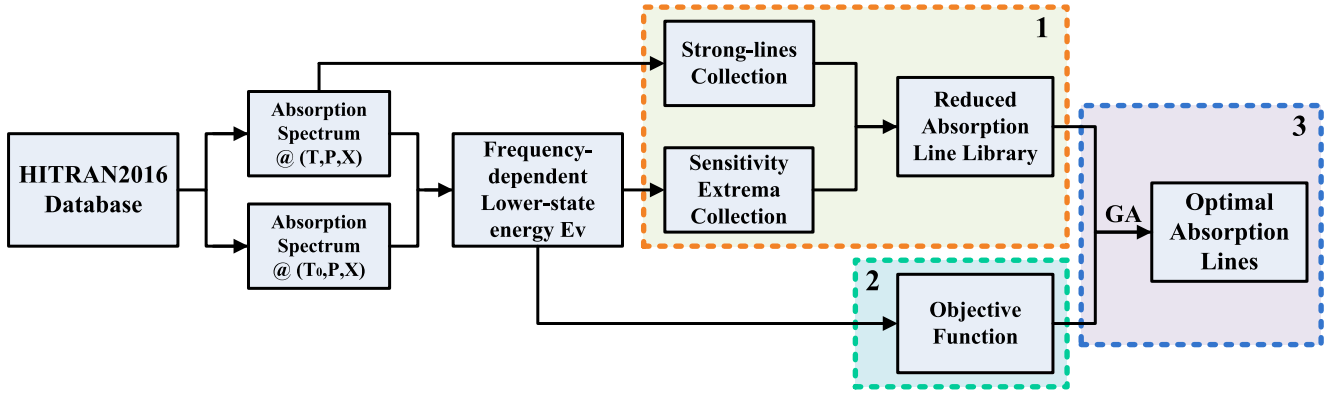
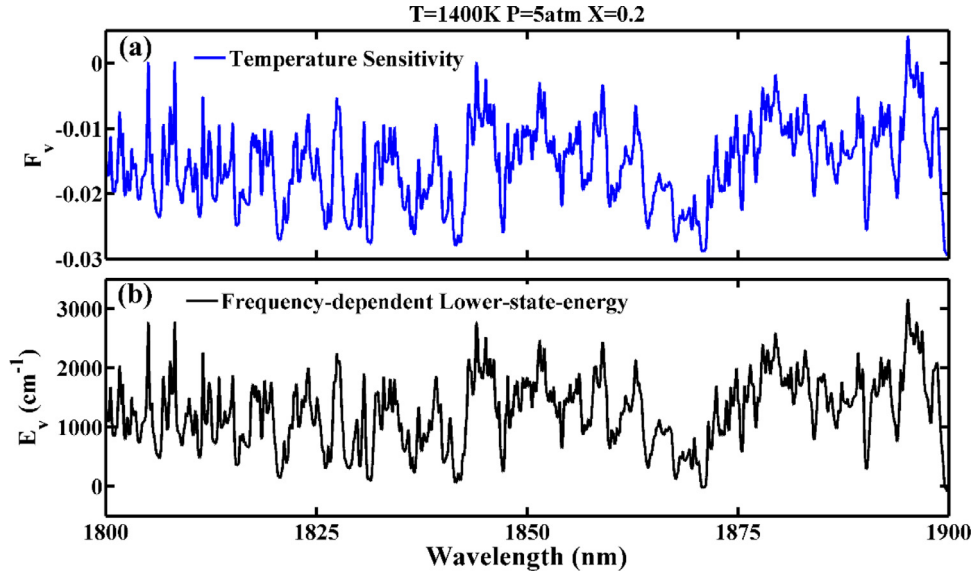


Fig. 3. Line-selection process for CARL method.

Fig. 4. Comparison between the frequency-dependent lower-state energy E_v and the temperature sensitivity factor F_v . (a) temperature sensitivity factor F_v ; (b) frequency-dependent lower-state energy E_v .

tion and the sensitivity extremes collection. The strong-lines collection is built by gathering the absorption peaks of the absorption spectrum. And, similarly, the limits of the E_v spectrum forms the sensitivity extremes collection. The intersection of these two collections gives the reduced absorption line library.

Compared to CA method, the sample space in CARL method can be remarkable reduced by several orders of magnitude. As a result, the line-picking step can be remarkably accelerated.

2.3. Simplified method with hot-line and cold-line libraries

To further simplify the selection process, instead of the frequency-dependent lower-state energy E_v , here, we define a temperature sensitivity factor F_v to measure the temperature sensitivity of different absorption lines. And, the temperature sensitivity factor F_v is expressed as the natural logarithm of the spectrum ratio,

$$F_v = \ln[R_{T,T_0}(v)] = \ln \frac{a(T, v)}{a(T_0, v)}. \quad (4)$$

Together with Eq. (1), we have

$$E_v = \frac{k_B}{hc(\frac{1}{T_0} - \frac{1}{T})} \left\{ F_v + \frac{1}{2} \ln \frac{T}{T_0} + \ln \frac{Q(T)}{Q(T_0)} \right\}, \quad (5)$$

$$= AF_v + B$$

where A and B are temperature dependent parameters.

As can be found in Eq. (5), for a given temperature, the frequency-dependent lower-state energy E_v and the temperature sensitivity factor F_v show a linear relationship. Therefore, they should share the same spectral profile, as illustrated in Fig. 4. Hence, the relatively complicated frequency-dependent lower-state energy E_v can be equated to the simple temperature sensitivity factor F_v .

With the temperature sensitivity quantified, the absorption lines in the reduced library can now be further classified into two groups, the hot-line group and the cold-line group. Then, the optimal lines can be directly picked from the two separate groups, making the wavelength selection even simpler. Since this simplified method exploits hot-line and cold-line libraries, here, it's named as the HC method. And, the line-selection process for the HC method is diagramed in Fig. 5.

In HC method, as can be seen, only two major steps are involved, generation of the libraries and line-picking. Here, the reduced library obtained in CARL method can be taken directly, as the temperature sensitivity factor F_v and the frequency-dependent lower-state energy E_v share the same spectral profile. Then, a demarcation is exploited to separate the absorption lines. Usually, the smooth fitting or the average value of the temperature sensitivity factor is taken as the critical line. Lines with temperature sensi-

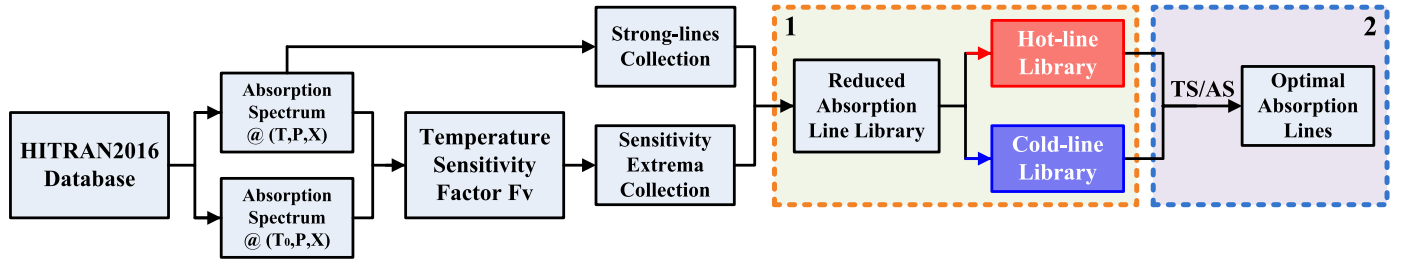


Fig. 5. Line-selection process for HC method. TS: temperature sensitivity approach; AS: absorption strength approach.

tivity factors above the critical line are marked as hot-lines, while lines with temperature sensitivity factors below the average are labeled as cold-lines.

In the line-picking step, the evolution algorithm based optimization process in CA and CARL methods is replaced with a much simpler approach. In this report, two line-picking mechanisms are exploited, namely the absorption strength approach (HCAS method) and the temperature sensitivity approach (HCTS method). For HCAS method, absorption lines are picked from libraries based on their absorption strengths. And, lines with top-ranked absorptions from the hot-line library and cold-line library are taken as the hot-lines and cold-lines, respectively. Differently, for HCTS method, absorption lines are picked based on their temperature sensitivity factors. Lines with the biggest temperature sensitivity factors from the hot-line library are chosen as the hot-lines and those with the smallest temperature sensitivity factors from the cold-line library as the cold-lines. For both approaches, the combination of the picked hot-lines and cold-lines forms the selection results. A crude theoretical analysis suggests that, HCTS method would probably give a better selection, as, by this means, lines with large temperature sensitivity differences are selected, which usually leads to a high temperature measurement precision.

Compared to CA method and CARL method, function optimization is no longer needed in HC method. Instead, the lines are di-

rectly selected from two small sample spaces based on their absorption strengths or temperature sensitivity factors, making the wavelength selection process much simpler, faster, and more direct.

3. Selection results and discussion

To validate the feasibility of the simplified methods, selection results for 2, 3 and 5 lines at different temperatures are presented and compared with those obtained with CA method. And, the temperature uncertainties for the simplified method are calculated for comparison.

3.1. Selected lines

3.1.1. CARL method

Typical selection results for 2, 3 and 5 lines with CARL method are depicted in Fig. 6. As stated above, for CARL method, the lines are selected from the reduced library, marked with small blue dots in Fig. 6. As can be seen, the absorption lines in the library hold both high temperature sensitivity and peak absorbance, which ensures a high temperature precision. For the investigated wavelength range from 1800 nm to 1900 nm, only 39 absorption lines are selected to form the reduced library, much smaller than the

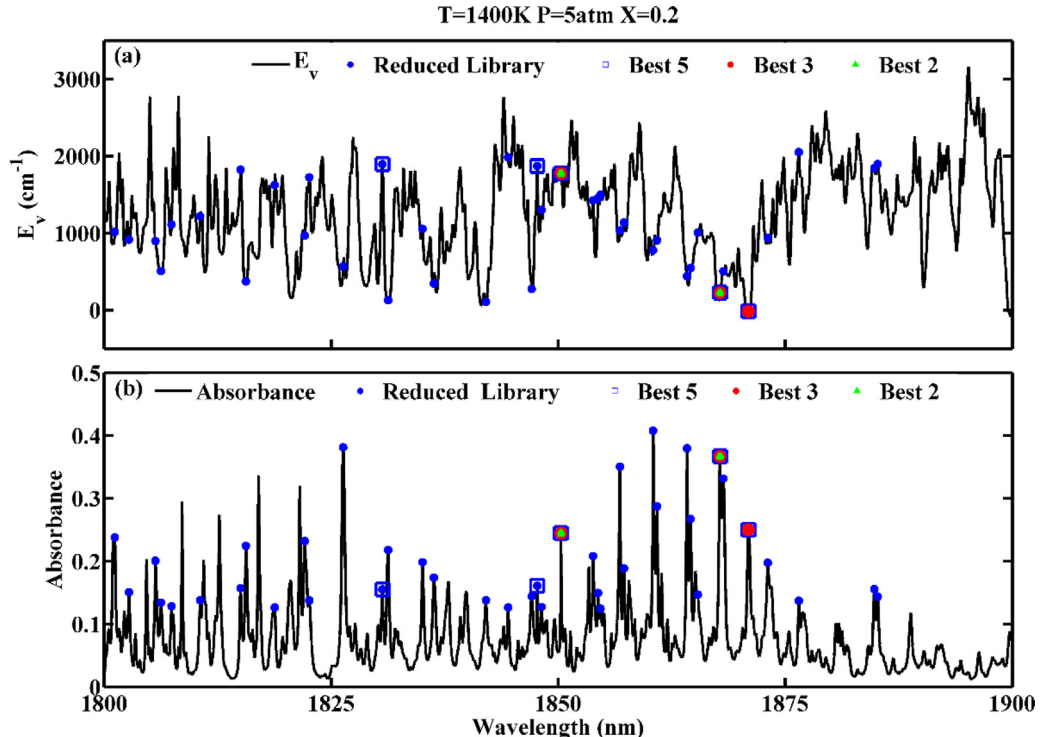


Fig. 6. Reduced library and best selection results for 2, 3 and 5 lines at 1400 K with CARL method. (a) frequency-dependent lower-state energy E_v ; (b) absorbance. The small blue dots, the blue square, the red circles and the green triangles represent the reduced library, the best 5 lines, the best 3 lines and the best 2 lines, respectively.

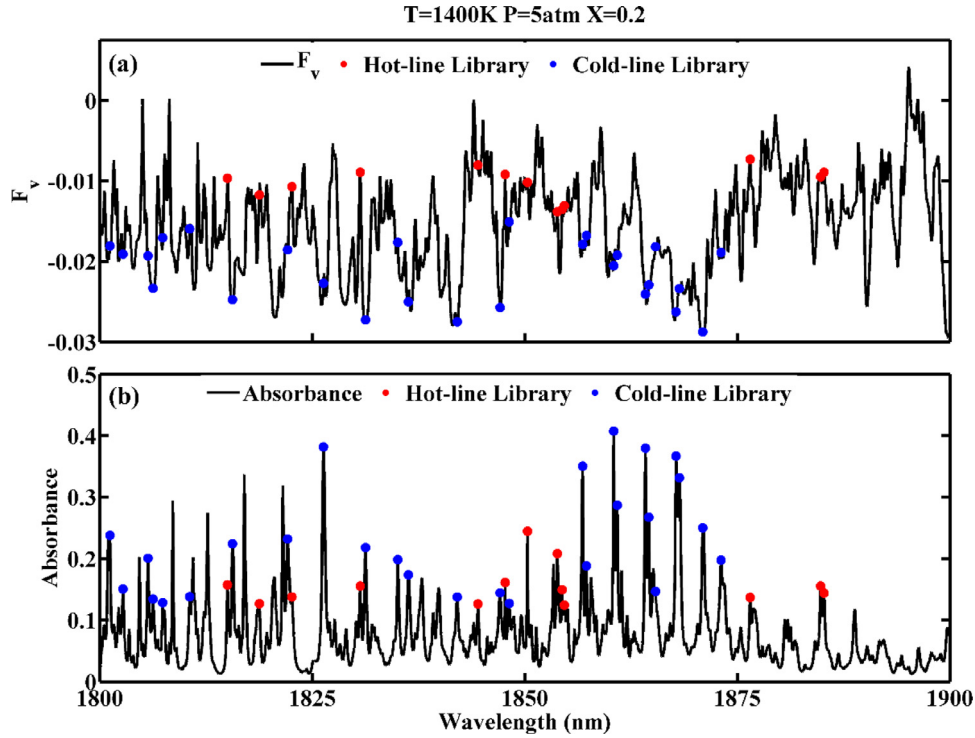


Fig. 7. Hot-line library and cold-line library at 1400 K. (a) temperature sensitivity factor F_v ; (b) absorbance. The red and the blue circles represent the hot-line library and cold-line library, respectively.

sample space of CA method which contains ~ 3000 lines. More importantly, comparisons with the lines in Fig. 2 show that, CARL method presents the same selection results with CA method, manifesting the feasibility of this method.

3.1.2. HCAS method and HCTS method

For HCAS method and HCTS method, the hot-line and cold-line libraries are established from the reduced library in Fig. 6. And, the average temperature sensitivity factor is taken as the classification criterion. Fig. 7 depicts the locations of generated hot-line library and the cold-line library.

With the hot-line library and the cold-line library established, the lines can now be picked based on their absorption strengths or temperature sensitivity factors, depending on the methods exploited. In HCAS and HCTS method, for a given total line number N ($N \geq 2$), M ($M \in [1, N - 1]$) hot-lines with $N - M$ cold-lines can be selected, giving $N - 1$ kinds of combinations featuring different numbers of hot-lines. Here, for simplicity, only 4 candidates from each library are provided, since the maximal line number is set to be 5. And, the selected results are given in Fig. 8. The candidate selection is quite simple. For HCAS method, the lines are ordered by their absorbance. And, top 4 lines from each libraries are taken as the candidates. For HCTS method, the lines are ordered by their temperature sensitivity factors. And, top 4 lines from the hot-line library are taken as the hot-line candidates, while, the last 4 lines from the cold-line library taken as the cold-line candidates.

The optimal lines are the combination of hot-lines and cold-lines from the candidates. And, the selection process is the same as that for the candidates. Here, we take the 3-line case as an example. For HCAS method, the selected 3 lines can be consisted of one hot-line with the biggest absorbance in the hot-line library and two cold-lines with the top 2 absorbance in the cold-line library, or otherwise, two hot-lines with the top 2 absorbance in the hot-line library and one cold-line with the strongest absorbance in the cold-line library.

Table 1

Computation cost of different line-selection methods.

Method	Whole line-selection process (s)			Line-picking step (s)		
	2-line	3-line	5-line	2-line	3-line	5-line
CA	85.35	122.57	265.73	82.53	119.36	256.29
CARL	20.10	25.55	44.20	11.05	14.90	35.73
HCAS&HCTS	9.17			0.003		

As can be found, with HCAS and HCTS methods, the magnitude-sorting based line-selection is more direct and much simpler. Besides, comparisons with the selection results in Fig. 2 demonstrate that the selected candidates cover the selected optimal lines by CA method.

3.1.3. Computation cost and line comparison

With reduced libraries, the simplified methods should present a much faster line-selection process. Table 1 lists the time cost by different line-selection methods. The selection operation is tested at a standard Lenovo P900 workstation. As can be found in Table 1, the CA method takes the longest time to get the optimal line combinations. And, over 95% of the time is consumed by the line-picking step. With a reduced library, the time cost of CARL method is remarkably shortened to about 1/5. Considering the extra cost caused by library filtering, the line-picking in CARL method is about 7~8 times faster than that in the conventional CA method. HCAS and HCTS methods demonstrate the highest efficiency with an overall line-selection time no more than 10 s, regardless of the line numbers. And, what's even more impressive is that, the line-picking step in HCAS and HCTS methods take only 0.003 s to find the optimal absorption lines, as the magnitude-sorting is an immediate process.

Fig. 9 presents the selected wavelength pairs by different selection methods at different temperatures. As can be seen, the lines selected by different methods vary from each other. And, the cold-

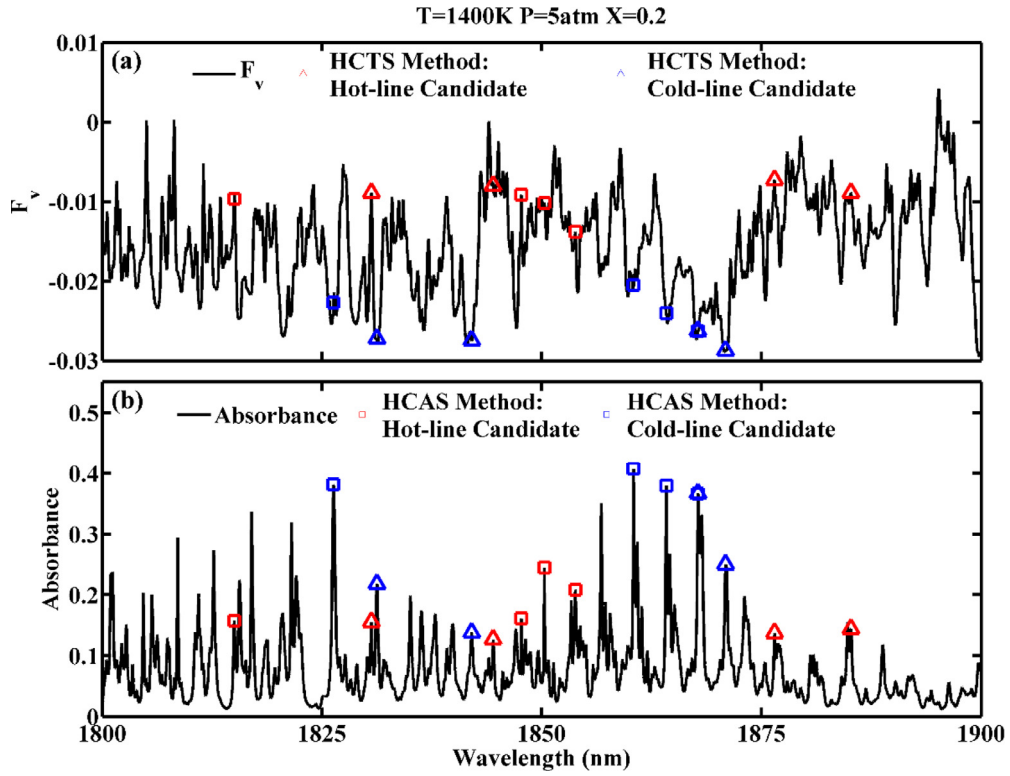


Fig. 8. Hot-line and cold-line candidates at 1400 K. (a) temperature sensitivity factor F_v ; (b) absorbance. The red and the blue represent the hot-line candidates and the cold-line candidates, respectively. And, the squares and the triangles represent the selected candidates by HCAS method and HCTS method, respectively.

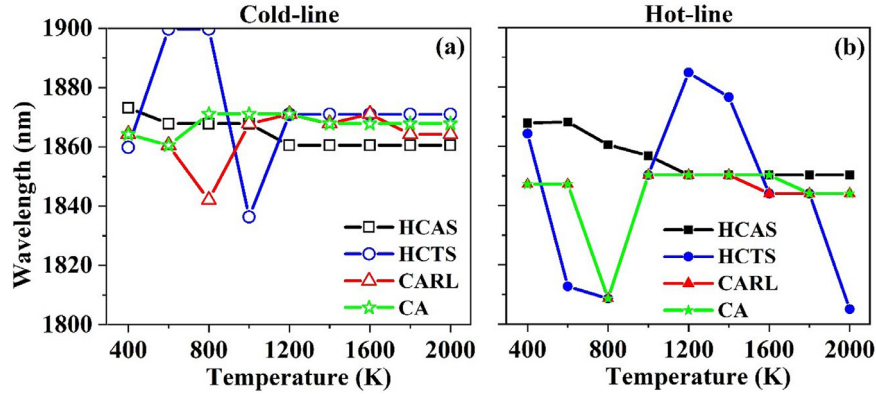


Fig. 9. Optimal wavelength pairs by different selection methods at different temperatures. (a) cold-line; (b) hot-line. The squares, circles, triangles and stars represent the lines selected with HCAS method, HCTS method, CARL method and CA method, respectively.

lines show more options compared with hot-lines. Generally, the selected lines of CARL method exhibit a high possibility to overlap with those given by CA method, especially for hot-lines. Lines selected by HCTS method show a wide spectral variation for different temperatures, while those by HCAS method are much more stable, especially at high temperatures.

Here, it should be noted that, the difference in wavelengths does not directly reflect the difference in measurement precision quantitatively, as there exist plenty of near-optimal wavelength pairs with comparable temperature precisions [30,31].

3.2. Selection evaluation

To give a more detailed evaluation of the different selection results, the temperature uncertainties of different selection methods are investigated. Fig. 10 compares the temperature uncertainties of different selection methods within a temperature range from 400 K

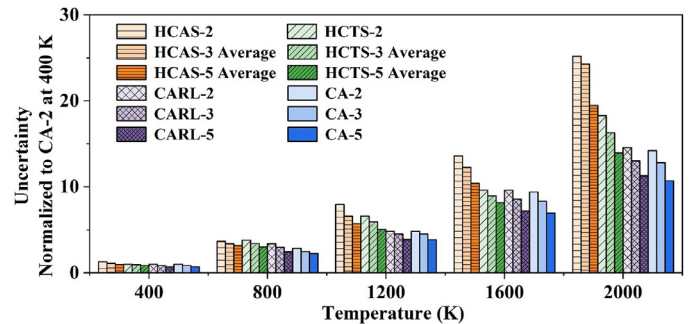


Fig. 10. Temperature uncertainties of different selection methods. HCAS/HCTS/CARL/CA-2/3/5: HCAS/HCTS/CARL/CA method with the best 2/3/5 absorption lines; HCAS/HCTS-3/5 Average: average uncertainty of HCAS/HCTS method with the best 3/5 absorption lines.

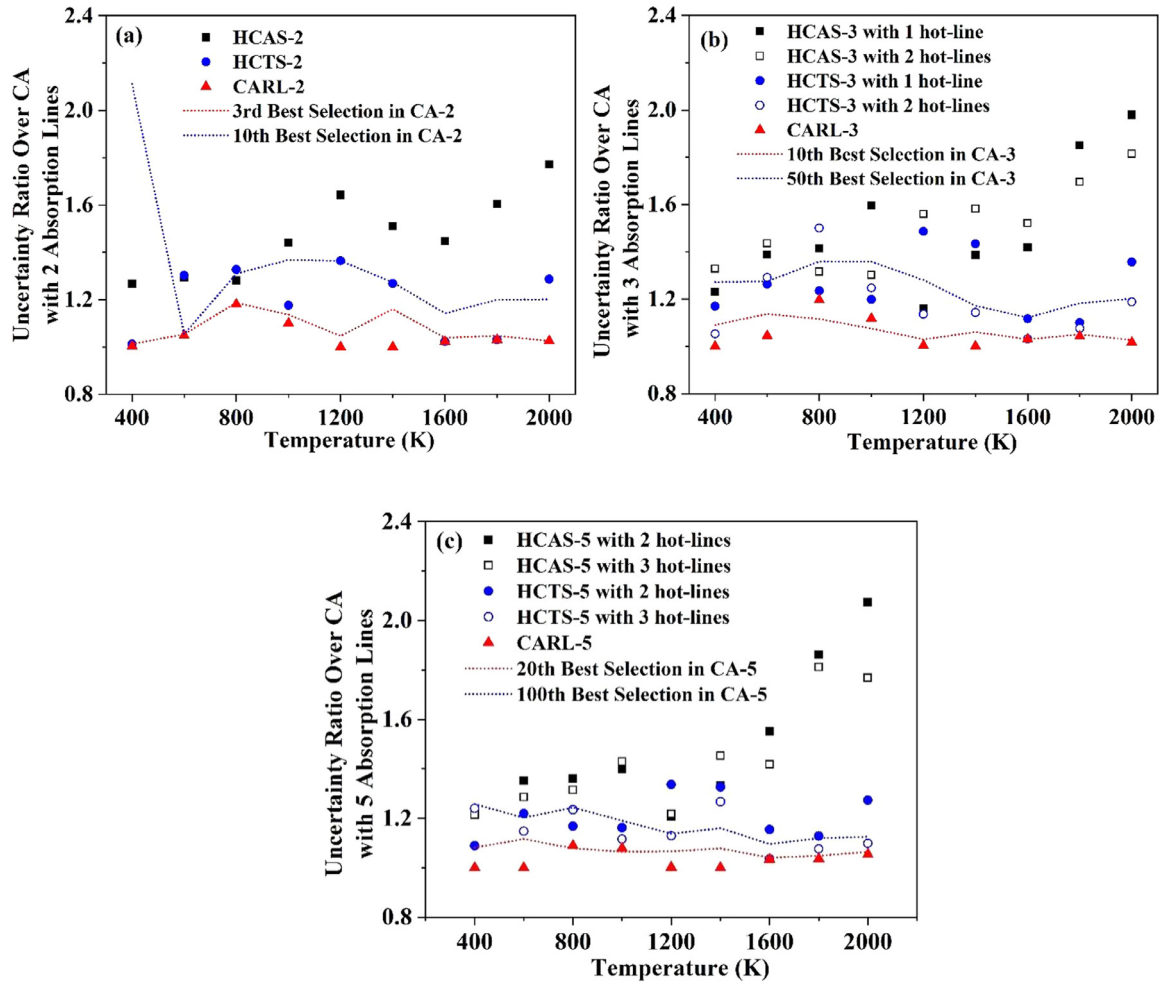


Fig. 11. Uncertainty ratios to the best selection results by CA method for different line numbers. (a) 2 absorption lines; (b) 3 absorption lines; (c) 5 absorption lines. The squares, circles and triangles represent the uncertainty ratios of HCAS method, HCTS method and CARL method, respectively. The dashed lines stand for the uncertainty ratios of the top 100 best lines selected by CA method.

to 2000 K. It should be noted that, for HCAS and HCTS methods with 3 and 5 lines, the average uncertainties are taken, as there exists more than one selection results. And, all the uncertainties are normalized to the best 2-line selection by CA method at 400 K.

As can be found in Fig. 10, overall, for different selection methods, the uncertainty rises with the temperature. This can be explained by the fact that the temperature uncertainty $\delta T/T$ is directly proportional to T , as given by Eq. (3). Secondly, for a given temperature, the temperature measurement precision improves with the number of absorption lines selected, which is consistent with the results in Ref. [32].

As for the three simplified methods proposed in this report, comparisons reveal that, CARL method exhibits the best performance with nearly the same precision with the traditional CA method, while HCAS method induces the largest uncertainties. A more clear view is provided in Fig. 11, where the detailed uncertainty ratios of different selection methods are plotted for different line numbers. Additionally, the relative temperature uncertainties of the top 100 best selections by CA method are also depicted for comparison.

It can be found in Fig. 11 that, CARL method exhibits the best temperature precision with uncertainty ratios over CA method kept within 1.2 for different line numbers throughout the investigated temperature range. For the cases of 2, 3 and 5 lines selection, CARL method enables a temperature precision comparable to the 3rd, 10th and 20th best selections in CA method, respectively. For HCTS

method, the maximal uncertainty ratio remains below 1.5. And, the precisions of the best selections are equivalent to the 10th, 50th and 100th best options in CA method for 2, 3 and 5 line cases. Generally, HCAS method displays a relatively larger temperature uncertainty than CARL method and HCTS method. Even so, with a maximal uncertainty ratio around 2 at high temperatures, the selection results of HCAS method are still tolerable in many applications [30].

4. Conclusion

Concerning the numerical line-selection of water vapor around 2 μm , the conventional CA method based on the frequency-dependent lower-state energy is simplified with a reduced library through pre-filtering of the candidate absorption lines. Besides, the temperature sensitivity factor is defined, replacing the frequency-dependent lower-state energy, to measure the temperature sensitivity of different absorption lines. Based on the reduced library and the quantified temperature sensitivity factor, three simplified line-selection methods are proposed. Of the three, the first method features a reduced library containing only tens of line candidates which is only about 1/100 that of the CA method, and provides the best performance. Compared to CA method, selection results with a maximal temperature uncertainty rise within 20% is obtained at a 7~8 times smaller computation cost. The other two methods are characterized by both a reduced library and a magnitude-

sorting based direct selection process. The direct selection is a much faster process realized by ranking the lines based on the absorption strength or the temperature sensitivity factor. Analysis shows that, the temperature sensitivity based method provides a better selection than the absorption strength based one. And, over the investigated temperature range, the maximal temperature uncertainty keeps below 1.5 of that attained by CA method. Nevertheless, even for the absorption strength based method, the uncertainty is still within the acceptable range.

These simplified line-selection methods are instructive for the line-selection in practical absorption thermometry applications, including both multi-discrete-line absorption and multi-band absorption. Besides, since the temperature sensitivity factor, just like the frequency-dependent lower-state energy, is a continuous quantity closely related to environmental conditions, the proposed methods should also be applicable for the scanning span selection of continuously tunable laser sources exploited in hyperspectral absorption based spectral applications.

Statements

The work described is original research that has not been published previously, and not under consideration for publication elsewhere.

The authors declare no conflicts of interests with other groups.

All authors have read this manuscript and agreed to submitted it to Journal of Quantitative Spectroscopy and Radiative Transfer.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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CRediT authorship contribution statement

Mengmeng Tao: Conceptualization, Methodology, Investigation, Formal analysis, Writing – review & editing. **Yamin Wang:** Data curation, Investigation, Writing – original draft. **Haolong Wu:** Data curation, Investigation, Writing – original draft. **Jingfeng Ye:** Funding acquisition, Writing – original draft. **Xisheng Ye:** Conceptualization, Writing – review & editing. **Weibiao Chen:** Supervision, Methodology, Writing – review & editing.

Data availability

Data will be made available on request.

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